

COSC 320: Assignment 5

This assignment is due **Friday, April 4 at 7 PM**. Late submissions will not be accepted. All the submission and formatting rules for Assignment 1 apply to this assignment as well.

1 List of names of group members (as listed on Canvas)

Provide the list here. This is worth 1 mark. Include student numbers as a secondary failsafe if you wish.

1. Rin Meng, 51940633
2. Mika Panagsagan 29679552
3. Kevin Zhang 10811057

2 Statement on collaboration and use of resources

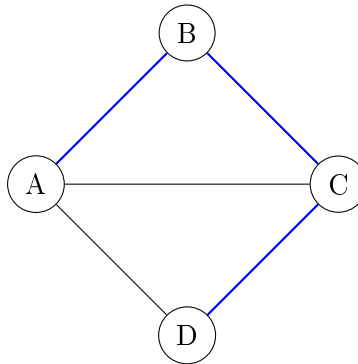
To develop good practices in doing homeworks, citing resources and acknowledging input from others, please complete the following. This question is worth 2 marks.

1. All group members have read and followed the guidelines for groupwork on assignments given in the Syllabus).
☒ Yes ☐ No
2. We used the following resources (list books, online sources, etc. that you consulted):
 - (a) DeepSeek R1, ChatGPT to generate ideas, explain concepts, and check if we are going in the right direction.
 - (b) Google Search, Google's AI Overview
 - (c) COSC 320 Lecture Slides
3. One or more of us consulted with course staff during office hours.
☐ Yes ☒ No
4. One or more of us collaborated with other COSC 320 students; none of us took written notes during our consultations and we took at least a half-hour break afterwards.
☐ Yes ☒ No
If yes, please list their name(s) here:
5. One or more of us collaborated with or consulted others outside of COSC 320; none of us took written notes during our consultations and we took at least a half-hour break afterwards.
☐ Yes ☒ No
If yes, please list their name(s) here:

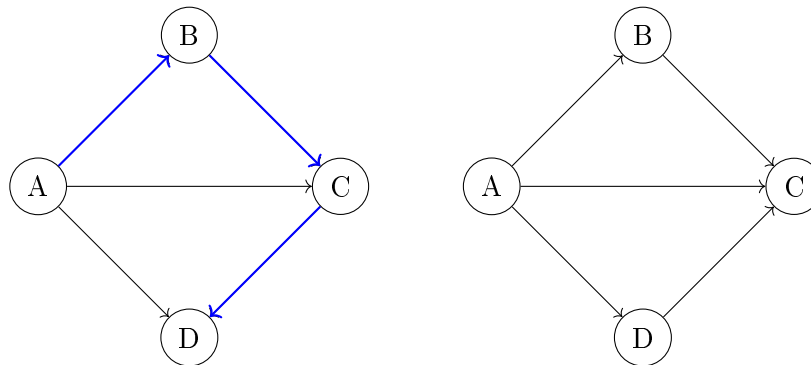
3 The Path To Victory 2.0

A Hamiltonian Path is a path in a graph that visits each vertex exactly once. In this question, we consider the variant of the Hamiltonian Path problem with the start and end node specified: that is, given a graph $G = (V, E)$ and two nodes s and t , does there exist a path from s to t that visits every node in G exactly once? You can may assume that G contains at least three vertices.

The Hamiltonian Path problem can be applied to graphs that are either undirected or directed. For example, the undirected graph below has a Hamiltonian Path from A to D given by (A, B, C, D), shown in blue:



For the directed graphs below, the graph on the left has a Hamiltonian Path from A to D, while the graph on the right does not have a Hamiltonian Path from A to D.



We refer to the undirected version of this problem as **UHP** and the directed version as **DHP**.

1. [3 points] Give a reduction from UHP to DHP.

Given an undirected graph $G = (V, E)$, we can construct a directed graph $G' = (V', E')$ as follows:

- For each vertex $v \in V$, create a vertex $v' \in V'$.
- For each edge $(u, v) \in E$, create two directed edges (u', v') and (v', u') in E' .
- Or more concisely, we have to make the edge $u \rightarrow v$ and $v \rightarrow u$ for all edges in the UHP, into DHP

Since every edge in the original graph becomes two directed edges (one in each direction), all the paths that existed in the undirected graph still exist in the directed one. In particular, any Hamiltonian Path from s to t in the undirected graph corresponds exactly to a Hamiltonian Path from s to t in the directed graph.

This implies that solving the problem in the directed graph will also solve the problem in the undirected graph.

2. [4 points] Prove the correctness of your reduction from UHP to DHP. That is, prove that the answer to your reduced DHP instance is YES if and only if the answer to UHP is YES.

To prove the correctness of the reduction, we want to show that;

If the original UHP instance is YES $\iff$ the reduced DHP instance is YES

Proof. $\Rightarrow$ **Direction: If UHP instance is YES**

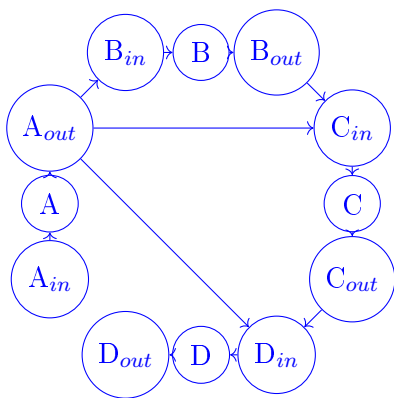
We assume first that the original UHP instance is YES. That implies that there exists a Hamiltonian Path from s to t in G that visits every vertex exactly once. Since G' contains all vertices of G and for every undirected edge from $(u,v) \in G$ we have both $(u,v) \wedge (v,u) \in G'$, we can also follow the exact same path as in G in G' . Therefore, we can find a Hamiltonian Path from s to t in G' that visits every vertex exactly once.

$\Leftarrow$ **Direction: If DHP instance is YES**

Now we assume that the reduced DHP instance is YES. That means that there exists a directed Hamiltonian Path. Since every edge in G' comes from a corresponding undirected edge in G , we can follow the same path in G as in G' . Therefore, we can find a Hamiltonian Path from s to t in G that visits every vertex exactly once. $\square$

3. [7 points] Give a reduction from DHP to UHP. You **do not** need to prove the correctness of this reduction, but you should clearly explain the key components of your reduction and why they are there.

To reduce from DHP to UHP, we first have to understand that directed graphs flows a certain way, that is, the edges are directed. In order to convert this “directional-ness” into an undirected graph, we can introduce two new vertices for every $v \in V$ such that the new vertices are v_{in} and v_{out} . So then the edges will look something like this



This gadget will allow us to convert the directed edges into undirected edges, by forcing it through a certain path, in this case, in the direction of $v_{in} \rightarrow v \rightarrow v_{out}$. So, for every directed edge $(u,v) \in E$, add the undirected edge $u_{out} - v_{in}$ to the undirected graph.

4 Strategically Placed Krispy Kremes

This SPKK problem is also in Tutorial 6. UBC Rec student leaders are planning their next fundraiser, and are seeking your help in identifying strategic locations to set up their stands of Krispy Kremes. They have a map showing n locations of buildings and outdoor spots on campus. Their k stands need to be set up in the outdoor spots. They want you to select k spots such that the maximum distance from any of the n locations to a stand is as small as possible.

An instance of the Strategically Placed Krispy Kremes decision problem (SPKK) has a set V of size n , a subset S of V , an integer k , $1 \leq k \leq |S|$, and a symmetric matrix $d[1..n][1..n]$, plus an additional nonnegative integer b . The problem is to determine if there is a subset $S' \subseteq S$ of size k , such that

$$\max_{v \in V} \min_{s \in S'} \{d[v][s] \mid s \in S'\} \leq b.$$

1. [8 points] Provide a reduction from the Dominating Set problem to SPKK. Briefly justify why the reduction is polynomial-time (one sentence is sufficient). Carefully explain why your reduction is correct (two paragraphs is sufficient).

In the Dominating Set problem, we are given a graph $G = (V, E)$ and an integer k , and we want to determine if there is a subset $D \subseteq V$ of size k such that every vertex in V is either in D or adjacent to a vertex in D .

So connecting back to our SPKK problem, we can see that we have a set V of locations, a subset $S \subseteq V$, where we can put our stands, and a distance matrix d . The goal is to choose k spots from S such that every location V is “close” (within distance b) to at least one of the stands. There is a technique we can use, and that is to convert the graph structure into the distance matrix so that the property if Dominating Set perfectly matches our condition that every vertex is within a distance b of a chosen stand.

Constructing our reduction:

1. Instance Transformation:

- Let the set V in the Dominating Set instance be both the set of vertices and the S where stand can be placed, (i.e. $S = V$).
- Then, we can define the matrix d as follows:
 - $d[u][v] = 0 \Leftrightarrow u = v$ (the distance from a vertex to itself is 0)
 - $d[u][v] = 1 \Leftrightarrow (u, v) \in E$ (the distance from a vertex to its neighbour is 1)
 - $d[u][v] = 2 \Leftrightarrow (u, v) \notin E$ (the distance from a vertex to its non-neighbour is 2)
- Then we set the bound $b = 1$, because we want to ensure that every vertex is either in D or adjacent to a vertex in D , (even if it is inefficient, at least it is better than non-polynomial time.)

2. Correctness of the reduction:

“If the graph has dominating set of size k ” Assume that we indeed have a dominating set $D \subseteq V$ with $|D| = k$. Then that means, for every vertex $v \in V$, either v is in D (so that $d[v][s] = 0$) or v is adjacent to a vertex in D (so that $d[v][s] = 1$). In either case, the maximum over all vertices of these minimum distances is at most 1.

Therefore, it must be true that, the maximum distance from any vertex to a stand is at most 1, which is less than or equal to b . Ultimately, implying that the SPPK instance with bound $b = 1$ is a YES instance.

“If the graph does not have a dominating set of size k ” Assume that we have a subset $S' \subseteq S$ with $|S'| = k$. Such assumptions implies that every vertex $v \in V$ is at most distance 1 from some $s \in S'$. This means for each v , either $v \in S'$ giving $d[v][v] = 0$ or there is some $s \in S'$ $d[v][s] = 1$, which implies that v and s are neighbours. Therefore, every vertex is either in S' or adjacent to a vertex in S' , so S' is a dominating set of size k .

Polynomial-Time Reduction Justification: The reduction is polynomial-time because the construction of the distance matrix d takes $O(n^2)$ time, where n is the number of vertices in the graph G . The construction of the distance matrix is a simple operation of iterating through the vertices and edges of the graph, and filling in the distance matrix accordingly. The other steps of the reduction,

such as defining the set S and the bound b , are also simple operations that can be done in polynomial time. Therefore, the overall reduction is polynomial-time.

5 Very Special Problems

In this question, we consider special cases of NP-complete problems. Formally, we say that Problem B is a **special case** of Problem A if every instance of Problem B can be viewed as an instance of Problem A. We've seen examples of this in class: for example, 3-SAT is a special case of SAT (where every clause has length 3). Minimum spanning tree is a special case of Steiner Tree, in which the set of vertices we need to connect includes all the vertices in V .

1. [4 points] Consider the **Bounded-Leaf Spanning Tree Problem (BLST)**: given a graph $G = (V, E)$ and an integer k , does G have a spanning tree with no more than k leaves?

Give an NP-complete problem that is a special case of BLST, and justify why this problem is a special case.

Hamiltonian Path is an NPC problem that is a special case of BLST. For the Hamiltonian Path problem, we are given a graph $G = (V, E)$ and we want to determine if there is a path from s to t that visits every vertex in G exactly once. For BLST, we are given a graph $G = (V, E)$ and an integer k , and we want to determine if there is a spanning tree with no more than k leaves.

Justification: In BLST, a leaf is a vertex with degree of 1 in the spanning tree, and the Hamiltonian Path problem can be viewed as a special case of BLST where $k = 2$, and Hamiltonian Path in G as a spanning tree with exactly 2 vertices of degree 1 (the start and end vertices of the path). Therefore, this corresponds to spanning tree with exactly two leaves.

In conclusion, Hamiltonian Path is equivalent to BLST with $k = 2$. Since Hamiltonian Path is NP-complete, and BLST is a special case of it,

2. [3 points] You showed in the previous question that there is an NP-complete problem that is a special case of BLST, and it is not difficult to show that BLST is in NP (though we are not asking you to do this). Does this imply that BLST is NP-complete? Justify your answer.

Yes, it does imply that BLST is NP-complete.

Special Case and NP-Hardness: In the previous part, we showed that the Hamiltonian Path problem (which is NP-complete) is a special case of BLST. This means that every instance of the Hamiltonian Path problem can be viewed as an instance of BLST (by setting $k = 2$ in BLST). Since the Hamiltonian Path problem is NP-complete, BLST must be at least as hard as the Hamiltonian Path problem. So it must be true that this implies that BLST is NP-hard.

BLST is in NP: We are told that BLST is in NP. When you have a problem that is NP-hard and also belongs to NP, the problem is NP-complete.

So, since BLST has an NP-complete special case (making it NP-hard) and it is also in NP, it follows by definition that BLST is NP-complete.

3. [4 points] Give an example of a polytime-solvable problem you have seen in this class that is a special case of an NP-complete problem. Justify your answer.

The **Minimum Spanning Tree (MST)** problem is a polytime-solvable problem that is a special case of the **Steiner Tree** problem.

Justification: In the MST problem, we are given a connected, undirected graph $G = (V, E)$ and we want to find a spanning tree with minimum weight. In the Steiner Tree problem, we are given a connected, undirected graph $G = (V, E)$ and a subset of vertices $T \subseteq V$, and we want to find a minimum-weight tree that spans all vertices in T .

The MST problem can be viewed as a special case of the Steiner Tree problem where $T = V$. In this case, the goal is to find a minimum-weight tree that spans all vertices in V , which is exactly what the MST problem asks for. Since the MST problem can be solved in polynomial time using algorithms like Prim's or Kruskal's algorithm, it follows that the Steiner Tree problem is at least as hard as the MST problem. Therefore, the MST problem is a special case of the Steiner Tree problem.